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Analysis of interpretive structural model of Indian railway security system by analytic hierarchy process (AHP)

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Abstract

Purpose – Security and safety have remained important concerns for mankind since ancient times. In the context of railways, however, the threat perceptions to safety and security have increased significantly lately. In view of this, the Indian Railways requires an effective and efficient security management system. The purpose of this paper is to propose an integrated approach to help develop the Indian railway security system (IRSS) by successively reducing the complexity of the system through a series of studies.

Design/methodology/approach – The relevant elements of the complex system of Indian Railways have been identified. The framework in which the elements exist and interact with each other has been clearly established using the interpretive structural modelling (ISM) technique. The output of ISM is further reduced in complexity by having different policy option profiles. A comparison of different option profiles has been done by a multi-criteria decision-making technique, the analytic hierarchy process (AHP), by choosing suitable criteria for comparison.

Findings – The following elements need to be pursued as the key objectives for making IRSS: protection of passengers, protection of property, modernisation, manpower enhancement, multi-skilling of staff, latest technology and enhanced legal powers.

Research limitations/implications – The present research can be extended in many important ways. Interpretive structural models for different contextual relationships can be developed and used for formulating and implementing customised security policy. Policy elements and the ISM structure obtained in this research can be utilised for the system dynamic modelling of IRSS. A pilot study can be done to implement the recommendations made in this study.

Practical implications – The ISM model developed can be implemented as a policy tool in enhancing the railway's security. Some of the policy elements proposed appear to be consistent with the strategic direction being undertaken in the railway security in the country.

Social implications – Security is an important concern for mankind and social civilisations. The results have significant welfare implications in India and the rest of the world.

Originality/value – The present study is one of the first approaches in a series of studies in railway security in India. The results of this study can be extended to other security scenarios with similar needs.

Keywords Interpretive structural modelling, Systems approach, Analytic hierarchy process, Indian railway security system

Paper type Research paper

The views expressed by the authors are based on a specific research study, and do not necessarily reflect those of Indian Railways Administration.

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1. Introduction

From early human civilisation, human beings have always adopted ways and means to secure themselves. Security refers to the various arrangements aiming for the protection or prevention against loss, theft or danger (Woellner and Dellancy, 1974). It also implies being free of exposure to danger and the mechanism of prevention of attacks (Stephenson, 1981). With the passage of time, the protection of sovereign interests of political states led to the establishment of military forces to secure respective countries from external aggression. However, the changing scenario of world politics since the 1980s, ethno-national conflicts, religious fundamentalism, growth of non-state actors, newer forms of terrorism and other such reasons have thrown in newer challenges for security agencies throughout the world (Bunker, 2005).

To quite an extent, international borders have not been able to prevent collaborations between terrorist groups choosing to work together on grounds of shared ideologies, practices, commitments, funding strategies, etc. Terrorism has become a major concern in the recent times, which is aided negatively by advances in technology (Weinman, 2006). In this regard, the threat perceptions are one of the highest for public transport services, particularly the road and rail transport systems.

In Indian Railways, physical safety is ensured through appropriate measures relating to design features of railway carriages, engines, railway tracks, signals, etc.; deployment of trained personnel; and by practicing and ensuring of stipulated safe operational procedures, whereas security, though related with safety, refers to dealing with the risk (or harm) due to intentional criminal acts. Because of the financial implications involved, ensuring the highest standards of security and protection from all kinds of possible threats remain a difficult challenge. However, reasonable security arrangements can significantly mitigate the various threat perceptions. This can be achieved by the constant upgradation of skills, better processes, better technology, sound planning, appropriate training and so on. Due to the increase in professionalism across a range of services, the expectations of rail users have also been rising. Understandably, all stakeholders expect a perceptible improvement in the situation and increase in the satisfaction level of rail users.

The Indian railway security system (IRSS) is a complex system having a complicated structure. It is complex not only by way of the organisational structure, but also in operational functions and multiple interactions and relations between different factors of the system. The enormous complexity of the system is adequately indicated by the significant parameters, such as route length of 64,600 km, 7,146 stations, 12,335 trains, 23m commuters per day and 2.67 m-tonnes of freight per day, making it one of most biggest transportation systems in the world (Indian Railways Statistical Publications, 2012). Ironically, however, its enormity also makes its vulnerability on the higher sides in the railway systems. Thus, first the complexity of the IRSS needs to be understood, and then possible reduction in complexity should be attempted to derive meaningful policy implications. Thus, possible solutions in the form of technological quick fixes may not adequately address the problems of IRSS.

Considering the complexity and vulnerability of the railway system, it is clear that ensuring efficient security arrangements is one of the most challenging problems for security professionals. For instance, the exercise undertaken by the security personnel on 13 November 2006 to check the passengers by using multiple metal detectors at Churchgate railway station of Mumbai, India, had to be abandoned within a few minutes to avoid chaos. Likewise, at most of the important railway stations, a large number of foot falls make proper frisking of passengers and scanning of luggage nearly impossible, especially during the peak hours.

The Government of India has attempted to upgrade manpower, skills and technological setup of the railways security agencies to meet various challenges from time to time (Report of Comptroller and Auditor General of India, 2011–2012). However, the need of the hour is a demonstrable change at the ground level and increases in the user satisfaction. The overall objective of the present research work is to propose a comprehensive approach to help develop

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the IRSS by successively reducing complexity of the system through a series of studies. While attempting to reduce the complexity of large scale systems, it is important that the process should follow the scientifically proven steps. After the relevant elements of a complex system are identified, it is imperative that the framework in which the elements exist and interact with each other should be clearly established using soft systems methods like interpretive structural modelling (ISM). Group decision making is an important aspect of soft systems method because different experts may have different opinions on relevant issues, while collectively they are likely to agree on various parameters (Kralj, 2009). ISM is a consensus building method. It leads to a graphical representation of the structure of the system concerned, with the elements as nodes and relations as edges (Warfield, 2006). The output of ISM is further reduced in complexity by having different policy option profiles. A comparison of different option profiles has been done by a multi-criteria decision-making (MCDM) technique, the analytic hierarchy process (AHP), by choosing suitable criteria for comparison.

The rest of this paper is organised as follows. The next section provides a brief overview of the relevant literature in this area. Section 3 discusses the research methodology followed in the current research work. Section 4 presents the results obtained in this work and the associated policy implications of the results. The last section concludes with discussion on future work in this area.

2. Background literature

A significant security event in the current century has been the September 11 attacks in the USA. Subsequently, the Government of USA took several unique initiatives at many fronts, including political, administrative, law enforcement, intelligence and financial to counterterrorism. Similar initiatives have also been undertaken in South Asia, such as Maoists insurgency in Nepal (Marks, 2008), women's role in Jammu and Kashmir (Sobhrajani, 2008), Sikh ethno-national movement (Kumar, 2008) and group conflicts in Sri Lanka (Peiris, 2008). *The Journal of International Security Affairs* focused part of an issue on the subject of terrorism. It covered topics such as Narco-terrorism (Ehrenfeld, 2002), the revolution in military affairs and America's New War against terrorism in the wake of September 2001 9/11 attacks (Blank, 2002). Other relevant international terrorism-related work appears in Weiner (2006).

The critical nature of international counterterrorism initiatives has been enunciated in the United Nations Security Council Resolution 1373 (2001), calling for greater international cooperation (US Department of State Report on Global War on Terrorism, 2001). As stated by Latimer (2004), a principal threat to global and national security is terrorism. A comprehensive overview of various issues pertaining to national security, such as terrorism in Kashmir, separatist movement in Punjab, terrorism in North-East, naxalism, communal riots, etc., appears in Saksena (2001). Railways systems worldwide remain quite vulnerable to terrorist attacks (Riley 2004). In fact, the ground transportation systems remain even more vulnerable than the aviation sector (Greiper and Sauter, 2006). This is also due primarily owing to the openness and accessibility of the system (CRS Report, 2006). The Department for Transport, UK has published a number of studies related to the terrorist attacks on railways in London city and their impact on commuter behaviour (Department for Transport Research Report, 2005).

Although previous researchers have attempted to study the complex scenario of security, still, there is a lack of scientific research in this area. Systems science can effectively bridge this gap by focusing on description, design, complexity and action (Warfield, 2006). The nominal group technique (NGT) and the ISM of systems science approach are usually used to organise the complexity of a system.

The ISM methodology is rooted in soft systems approach, which provides significant inputs for policy making. It converts unclear, poorly articulated and

complex mental structure of systems into clear, simple and well-explained models. Starting from the 1970s, the ISM technique has been applied to a variety of settings such as fishery, planning, forestry and administration, across the world (Warfield, 1990). In India, starting from the seminal work at IIT Delhi, ISM approach has been used extensively. Agnihotri and Satsangi (1993a, b) developed a systems engineering framework for policy design for the Government of India for the handloom sector. More recently, the ISM methodology has been applied to diverse fields such as literature (Dayal and Satsangi, 2008), information risk management in supply chains (Faisal *et al.*, 2007), creation of flexible and lean value chain using outsourcing (Mohammed *et al.*, 2008), cement industry (Saxena *et al.*, 2006) and electrical/electronics industry's green supply chain (Govindan *et al.*, 2013). Srivastava *et al.* (2018) project futuristic scenarios for the IRSSs.

The AHP is a MCDM method, starting from the seminal work by Saaty (1972). AHP simplifies decision making when many conflicting and subjective decision criteria are involved (Saaty, 1977), and includes the following four steps: modelling, valuation, aggregation and sensitivity analysis (Ishizaka and Labib, 2009). The AHP has been widely implemented in banks (e.g. Haghighi *et al.*, 2010), manufacturing systems (Yang *et al.*, 2009), drugs selection (Vidal *et al.*, 2010), site selection (Önüt *et al.*, 2010), software evaluation (Chang *et al.*, 2009), evaluation of website performance (Liu and Chen, 2009), strategy selection (Chen and Wang, 2010; Wu *et al.*, 2009), supplier selection (Chamodrakas *et al.*, 2010), etc. A recent study of the use of AHP specifically in the context of the security assessment of high-speed railways appears in Setola *et al.* (2017).

3. Methodology

The ISM method converts the mental model of a complex system into a better articulated representation in graphical form. A pair-wise comparison of elements of the system by the experts leads to structural self-interaction matrix, which is then converted into a graphical representation. The resulting depiction, known as diagraph, provides the hierarchy between the elements of the system.

Usually, the intermediate steps involving sequential model exchange isomorphism are circumvented by using appropriate software. In the current work, we obtained the software from Professor John A. Warfield through Professor Benjamin Broom of George Mason University. The software requires feeding of the details of elements and the triggering question defining a relationship amongst them. Once all the questions are answered, it generates the level sets of matrices for drawing the structure of the system concerned. Some software are programmed to generate the final diagraph also.

For policy development purposes, the structure derived from ISM can be further subjected to option profile field and option profile methodology to develop different option profiles for enabling the policy makers to take a well-considered decision. Option fields can be obtained from the ISM of the entire set of relevant elements as a limited set of elements and dimensions considered necessary to achieve a given objective of the system. The experts are asked to ensure proper categorisation of options (elements) into dimensions, if not already done. Then they check the dependencies amongst the dimensions and cluster the dimensions accordingly. Further, the dimensions are sequenced on the basis of their requirement for consideration in choosing options for developing policy designs. The elements are also sequenced within each cluster in the same manner.

Option profiles are derived from the relevant option fields by asking the experts to choose options from each cluster and dimension to meet a given objective. The option profiles can be compared using the fuzzy technique or AHP.

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3.1 Process

After successfully reducing the complexity of IRSS during content analysis considerably at the element and dimension levels, it was subjected to ISM and option field and option profile modelling (OF&OPM) in this study in two phases. In the first phase, the ISM technique was utilised for structuring the elements finalised after content analysis. In the second phase, the OF&OPM method was utilised to develop three different option profiles. The AHP was used to compare the option profiles to assess their relative usefulness for IRSS.

3.2 Phase 1: ISM

The elements identified through NGT were refined further and input in the ISM software. The concept of ISM and its application in the railway security was then explained. Further, the ISM activity was carried out in a workshop involving pair-wise comparison of the elements. If there was a difference of opinion amongst participating experts, we held discussions so that a consensus could emerge. The participants were then shown the final matrix and the hierarchical structure obtained.

3.3 Phase 2: OF&OPM

In a workshop conducted to apply option profile field and option profile methodology for reducing complexity to a further extent, the detailed structure obtained through the ISM technique and fine-tuned by 15 domain experts was displayed for better appreciation by the experts. They were asked to briefly review the work done during content analysis and ISM. As the next step, the experts were asked to discover the clusters of dimensions by responding to the relational question: "Is dimension X independent of Y?" Further, in order to discover the sequential importance of the clusters and independent dimensions, the experts responded to the relational question: "Would it be better to consider Cluster/Independent Dimension I before Cluster/Independent Dimension J in choosing options for policy design?" Similarly, the dimensions were sequenced within each cluster by asking the participants to respond to the relational question: "Would it be better to consider Dimension N before Dimension M in choosing options for policy design?"

Subsequently, the experts were asked to identify the objectives and choose policy options for each of the objectives, selecting at least one option from each dimension contained in the option field so as to ensure that none of the important aspects are left out. This gave rise to three option profiles. In a subsequent workshop, the alternative profiles were compared by using the AHP method. The results obtained were processed using the Expert Choice software. The software required feeding in of the responses on goal specific criteria and the alternatives. The results were presented to the participants, and they were asked to discuss and deliberate to synthesise an even better option profile by consensus.

4. Results and analysis

4.1 Interpretive structural modelling

After the ISM exercise, the elements were finally structured into three main overall groups as depicted in Figure 1. The detailed ISM with coded elements is shown in Figure 2.

The five objectives shown in Table I emerged in cycles (Figure 3) at the top-most two levels.

As an objective, it was found that NRS10 (passenger-property protection) was of great importance. But one cannot ensure the protection of passengers and their property/railway property without ensuring the protection of human rights. As per the experts' opinion, the two objectives can be combined together for a better representation. The other three elements (security modernisation, secure environment for development and sea-life security) appeared in a cycle. This was owing to the fact that when terrorists possess the

sophisticated technology, one cannot neglect the importance of modernisation of railway security infrastructure. Sea-life or maritime security is also important, as exemplified by the attack on Chhatrapati Shivaji Terminus of Mumbai, India on 26 November 2008, which was carried out by entering the country through the sea route (*The Telegraph*, 2014). Thus, it can be inferred that we need a secure environment for the development and smooth functioning of railways.

4.2 Option field and option profile modelling

The option field development led to identification of three clusters and two independent dimensions is shown in Table II.

The prioritised sequencing of clusters and independent dimensions as well as elements within each dimension is shown in Table III.

The option profiles obtained are shown in Tables IV–VI, and ISMs associated with them are shown in Figures 4–6, respectively. The salient features of the profiles are discussed below.

4.2.1 Alternative option profiles. The experts identified the following three cases that could be considered as alternatives for policy formulation depending upon the desired status of IRSS.

Case 1: objective: passenger-property protection. As the title suggests, the main aim of this option profile is to provide security to passengers and their property. Funds for operational requirements, augmentation of the strength of Railway Protection Force (RPF), strengthening of coordination between intelligence agencies and legal empowerment of RPF are the basic requirements in this profile. These basic inputs are expected to ensure provision of the required electronic gadgets, proper registration and investigation of crime, and strengthening of the coordination of RPF with various states of the Union of India, which, in turn, would lead to the efficient maintenance of law and order in railway premises. Legal empowerment of RPF is also expected to ensure grievance redressal leading to the better protection of passengers and their property.

Case 2: objective: security modernisation. This option profile supports the premise that the purpose of securing railway system can be achieved by modernising the security setup. Apart from funds for operational requirements and augmentation of the strength of RPF, the basic elements required for achieving this objective include the prevention of cross-border intrusion and avoiding of bureaucracy in addition to laying a great emphasis on the collection of intelligence, its dissemination and strengthening of the intelligence setup. Significantly, legal empowerment does not figure as a basic requirement. While equipping RPF is expected to bring about some degrees of automation, yet, in this profile, office automation based on various aspects of intelligence figures as a prominent action for modernising security. Further, action for dealing with terrorism is considered important in this profile.

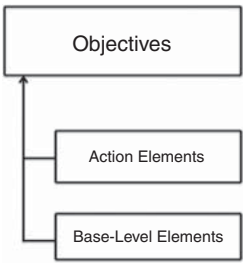


Figure 1.
Overall level ISM of Indian railway security system

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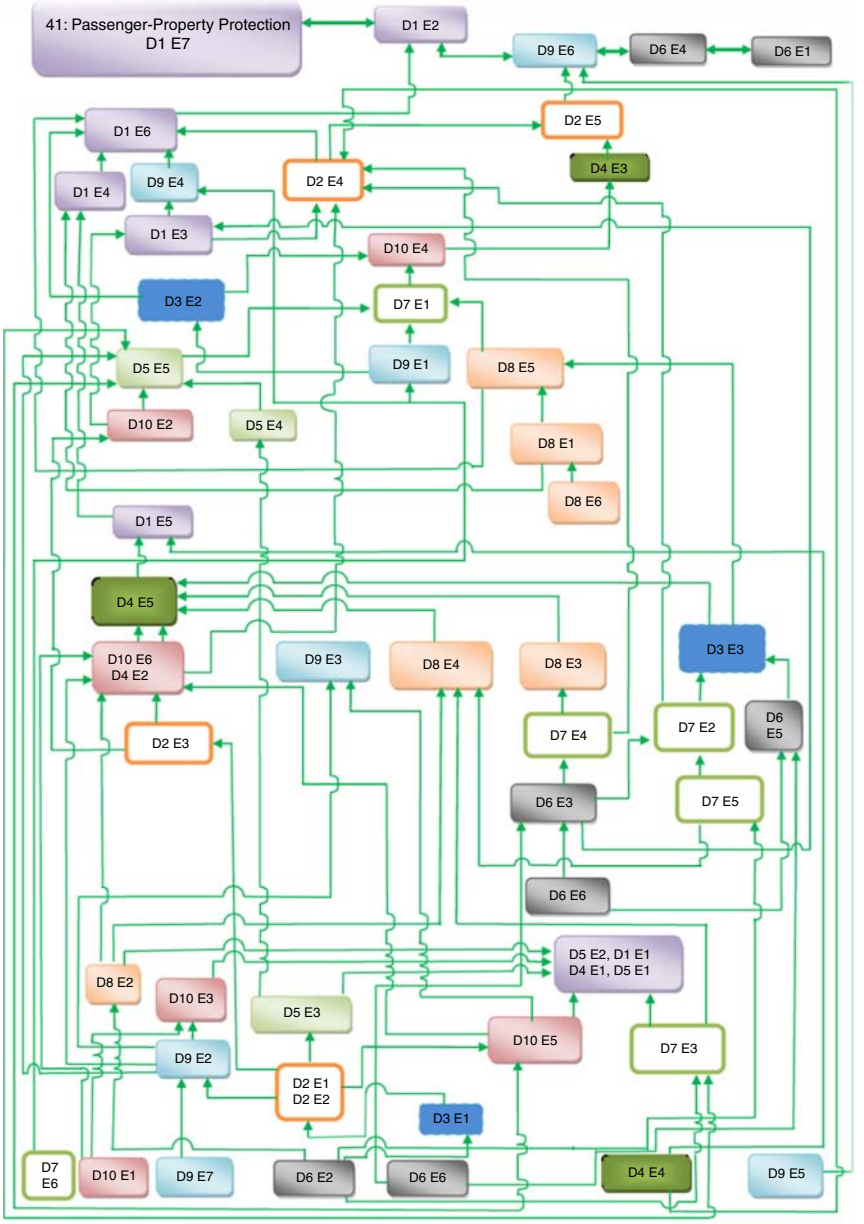


Figure 2.
ISM for Indian railway security system

Case 3: objective: passenger-property protection and security modernisation. Since the first two objectives are taken together as the objective in this option profile, most of the elements that figure here are present either in the first or the second option profile. The only differences are that ensuring a mechanism for redressal of grievances is not considered important, while boosting of public confidence is accorded significant place in the hierarchy of elements.

4.2.2 AHP analysis. In order to apply the AHP technique to compare the three different profiles, the problem was structured in terms of the goal (best possible performance of IRSS), criteria or features (feasibility, effectiveness, overall cost, reliability, technological advancement, scalability and bureaucratic and political interference) and alternatives

Table I. The top-level objectives in cycles (S. Nos. 1 and 2 in one cycle and S. Nos. 3–5 in another cycle)

S. No.	Code	Brief description	Key term	Dimension
1	NRS10	To protect railway property, passengers and passenger areas	Protect property, passenger, etc.	D1
2	NS4	To protect human rights of all citizens	Human rights protection	D1
3	NTAS17	To modernise security system, especially in the transport sector	Security modernisation	D6
4	NS10	To provide secure environment for development and economic growth	Provide secure environment for development	D6
5	NTAS7	To initiate measures for safety and security of life and property at sea and to take effective steps to ensure security against air borne attacks and chemical, biological, radiological, nuclear and electronic (CBRNE) attacks	Security at sea/Security against air borne attacks and CBRNE attacks	D9

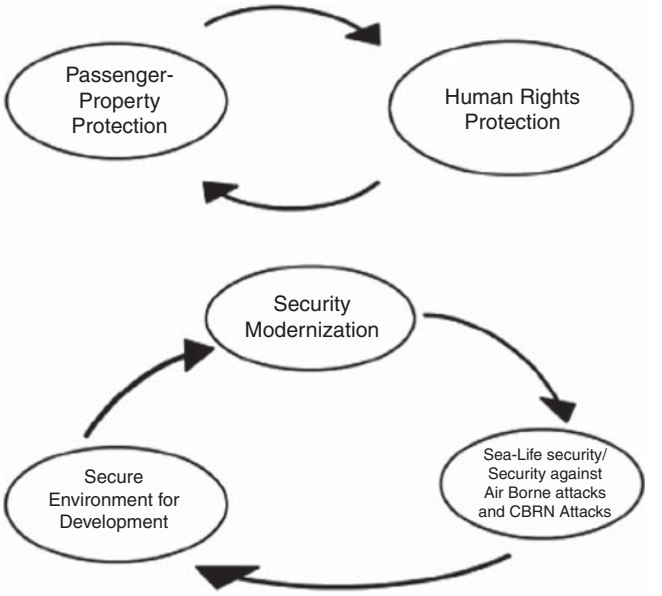


Figure 3. Cyclical depiction of top-level elements (objectives) for Indian railway security system

Table II. Clusters of dimensions and independent dimensions

Cluster 1 (C1)	D8 (human resources and infrastructure development), D7 (training, R&D and education), D3 (modernisation)
Cluster 2 (C2)	D1 (modernisation), D9 (legal aspects), D10 (administrative and office procedure)
Cluster 3 (C3)	D4 (inter-state security), D5(inter-border security)
Independent dimension (I1)	D6 (economy and finance)
Independent dimension (I2)	D2 (collection and dissemination of intelligence)

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(Case 1, i.e., Option Profile 1; Case 2, i.e., Option Profile 2; and Case 3, i.e., Option Profile 3) (refer Figure 7). The results obtained as response to the AHP questions were fed into the Expert Choice software. The results generated by the software are shown in Tables VII–XVI.

The first point to be noted from the AHP exercise is that the consistency ratios of different criteria while comparing the alternative option profiles and the consistency ratio for pair-wise comparison of different criteria with each other (0.08) are all less than 0.1 (refer Tables VII–XIV and XVI). Therefore, the results are consistent. Expert Choice varies the weights of criteria for sensitivity analysis. It is noted from the sensitivity analysis results that the priorities do not change.

The results also show that feasibility is considered as the most important criterion, followed by effectiveness, overall cost, reliability, technological advancement,

S. No.	Cluster/Independent dimension (in order of priority)	Dimension (in order of priority)
1	C1	D8 (human resources and infrastructure development), D7 (training, R&D and education), D3 (modernisation)
1(a)	I1	D6 (economy and finance)
3	I2	D2 (collection and dissemination of intelligence)
4	C2	D1 (social and cultural security), D9 (legal aspects), D10 (administrative and office procedure)
5	C3	D4 (inter-state security), D5 (inter-border security)

Table III. Elements within each dimension

S. No.	Cluster	Elements
1	C1	D8 E2, D3 E3
1(a)	I1	D2 E3
3	I2	D6 E5
4	C2	D10 E6, D9 E7
5	C3	D4 E2, D4 E5

Table IV. Objective 1 (passenger-property protection)

S. No.	Cluster	Elements
1	C1	D8 E2, D3 E3
1(a)	I1	D6 E5
3	I2	D2 E1, D2 E3
4	C2	D10 E2, D10 E1
5	C3	D4 E1, D5 E3, D5 E2

Table V. Objective 2 (security modernisation)

S. No.	Cluster	Elements
1	C1	D8 E2, D3 E3
1(a)	I1	D6 E5
3	I2	D2 E2, D2 E3
4	C2	D9 E7, D1 E6
5	C3	D4 E2, D4 E5

Table VI. Objective 3 (property protection and security modernisation)

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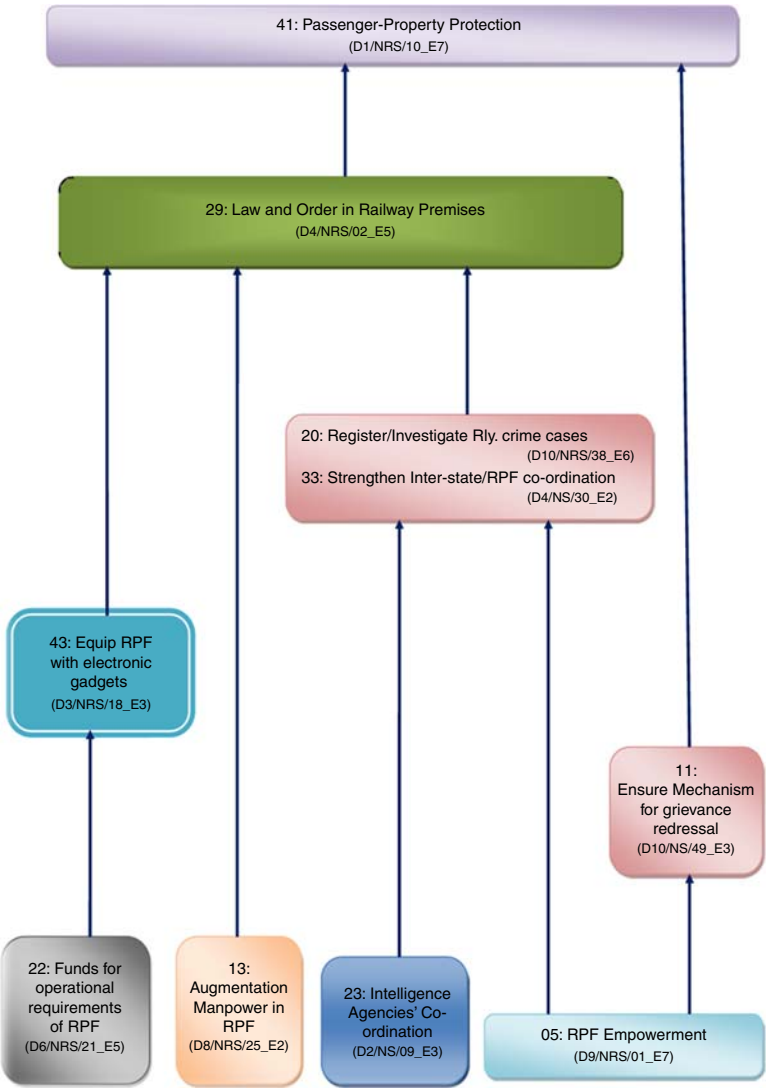


Figure 4.
Passenger-property protection

scalability and bureaucratic and political interference in that order. Overall cost at the third position was debated amongst the participants because usually financial constraints are considered to be a very high deterrent factor in enhancing security arrangements. However, finally, consensus was reached on the view that: even if every single rupee required for security is made available, it may not be feasible to provide the required technology because there is a need for research to develop suitable technology and that money cannot buy what does not even exist, and providing huge amount of money may not be feasible because it is likely to be impractical to spend more on security than the system can afford, thereby making the main business/service of transportation secondary to security.

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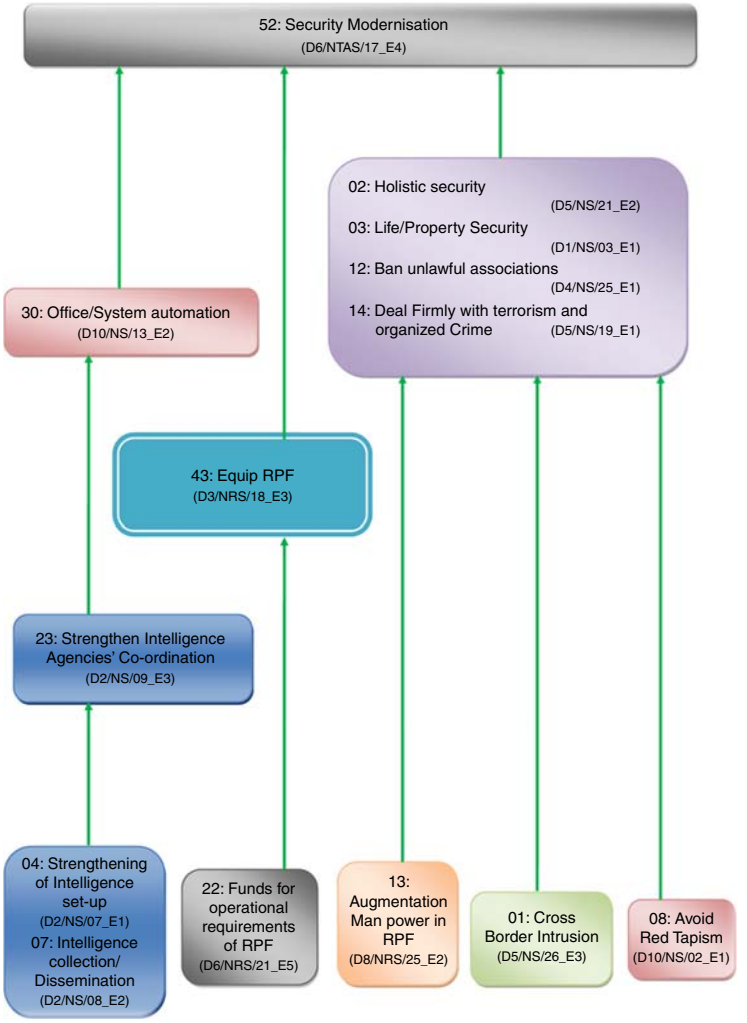


Figure 5.
Security modernisation

Further, the AHP exercise has revealed that Case 3 (Option Profile 3: life-property security and modernisation) is clearly the preferred profile. It is, however, a relatively expensive option because it lays emphasis on augmenting manpower, which is impractical and not feasible beyond a limit. The experts were of the view that augmenting the capabilities of existing manpower is a better and more feasible option. Therefore, the experts deliberated and wished to modify the profile to the extent that, instead of emphasising increase in manpower, capacity building of the existing force must be emphasised. This may not only be achieved by the required training but also by placing the latest technological options at their command. Accordingly, another option profile was synthesised as shown in Figure 8.

5. Conclusion and directions for future research

The pertinent literature review has revealed that there are gaps in the existing security system of Indian Railways. This is corroborated by the Report of the Comptroller and

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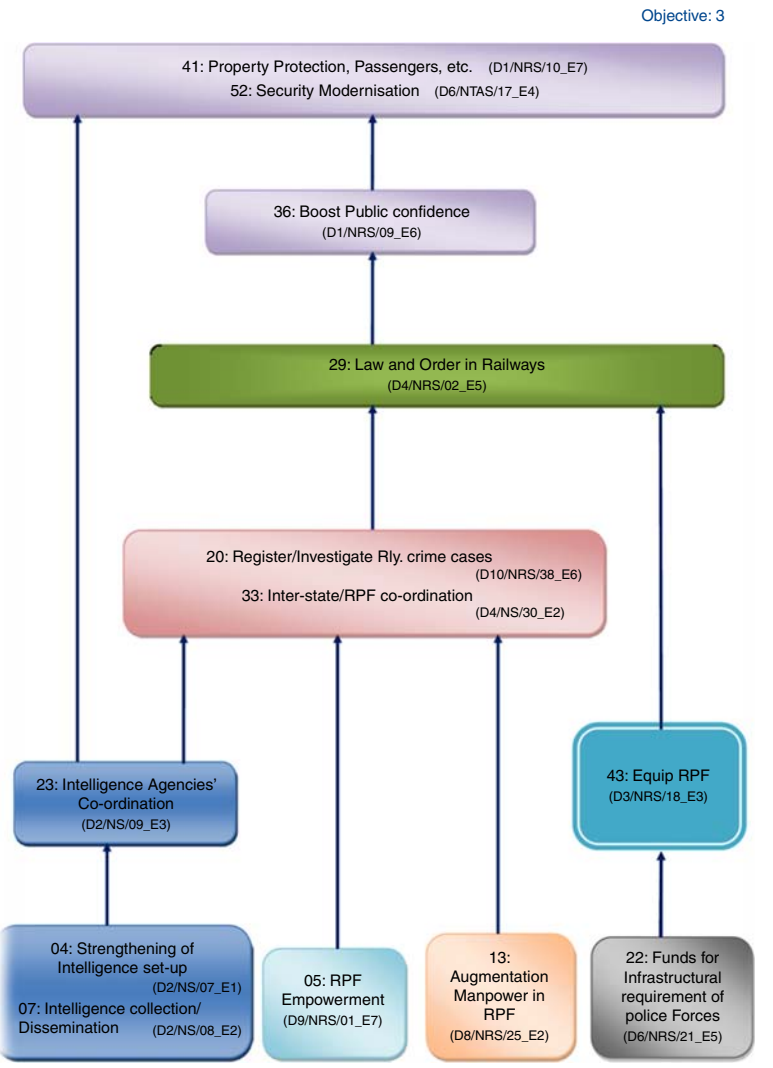


Figure 6.
Passenger-property protection and security modernisation

Auditor General of India (2011–2012), which has highlighted some of the gaps. The report itself says that no security system can prevent terrorist violence at public places completely, but good security arrangements can make the task of terrorists more difficult and improve the chances of detection.

In this context, it is necessary to strengthen the security arrangements in the Indian Railways to a level that commission of crime on railways does not remain an easy proposition for anti-social elements. Obviously, a lot is being done by the railway administration and the state governments to mitigate the problem (official website of RPF). Yet, as highlighted by the report of CAG for the year 2011–2012, there is need for a paradigm shift in dealing with this issue. However, no paradigm shift or a drastic change in security arrangement is possible in a complex system like that of Indian Railways without identifying various elements of the security system of railways and studying their interconnections.

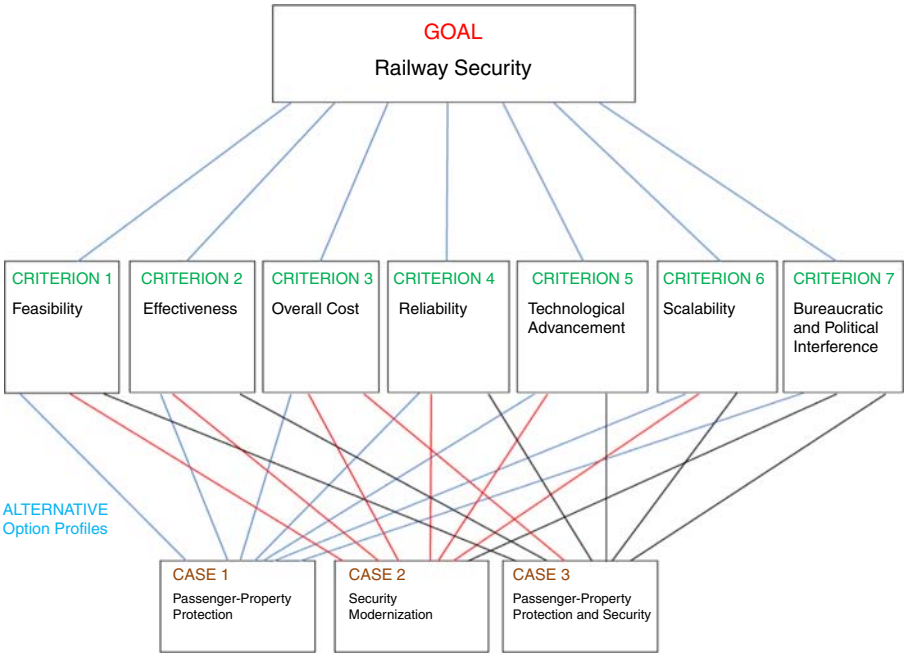


Figure 7.
Analytic hierarchy process diagram for Indian railway security system

	Feasibility	Effectiveness	Overall cost	Reliability	Technological advancement	Scalability	Bureaucratic and political interference	Priority
Feasibility	1	2	1	1	5	6	7	0.265
Effectiveness	0.5	1	2	3	4	5	6	0.253
Overall cost	1	0.5	1	2	3	4	4	0.187
Reliability	1	0.333	0.5	1	2	3	4	0.134
Technological advancement	0.2	0.25	0.333	0.5	1	2	3	0.069
Scalability	0.167	0.2	0.25	0.333	0.5	1	7	0.064
Bureaucratic and political interference	0.143	0.167	0.25	0.25	0.333	0.143	1	0.029
Consistency ratio	0.08							

Table VII.
Feature × feature matrix

	Case 1: Objective 1 (passenger security)	Case 2: Objective 2 (modernisation)	Case 3: Objective 3 (passenger security and modernisation)	Priority
Case 1: Objective 1 (passenger security)	1	0.5	0.333	0.163
Case 2: Objective 2 (modernisation)	2	1	0.5	0.297
Case 3: Objective 3 (passenger security and modernisation)	3	2	1	0.54
Consistency ratio	0.008			

Table VIII.
Comparison of alternative options on Criterion 1 (feasibility)

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In other words, proper policy planning would require reduction in the conceptual complexity of the system, preferably by applying the tools and methodology of systems science. A number of examples exist to prove that the application of system science tools and methodology has been utilised successfully to deal with complex social systems across the world (Warfield, 2006).

ISM for IRSS and the alternative profiles obtained have simplified the complex picture of IRSS considerably. They present a clear transitive relationship between the elements and are of great significance for policy making. As inferred from the results obtained, both the protection of passengers and property in the railway system, and modernisation are the objectives to be pursued for making IRSS effective and efficient. Modernisation has been viewed by the experts as almost synonymous with efficient status of security arrangements.

Table IX. Comparison of alternative options on Criterion 2 (effectiveness)

	Case 1: Objective 1 (passenger security)	Case 2: Objective 2 (modernisation)	Case 3: Objective 3 (passenger security and modernisation)	Priority
Case 1: Objective 1 (passenger security)	1	0.333	0.25	0.122
Case 2: Objective 2 (modernisation)	3	1	0.5	0.32
Case 3: Objective 3 (passenger security and modernisation)	4	2	1	0.558
Consistency ratio	0.016			

Table X. Comparison of alternative options on Criterion 3 (overall cost)

	Case 1: Objective 1 (passenger security)	Case 2: Objective 2 (modernisation)	Case 3: Objective 3 (passenger security and modernisation)	Priority
Case 1: Objective 1 (passenger security)	1	4	5	0.683
Case 2: Objective 2 (modernisation)	0.25	1	2	0.2
Case 3: Objective 3 (passenger security and modernisation)	0.2	0.5	1	0.117
Consistency ratio	0.021			

Table XI. Comparison of alternative options on Criterion 4 (reliability)

	Case 1: Objective 1 (passenger security)	Case 2: Objective 2 (modernisation)	Case 3: Objective 3 (passenger security and modernisation)	Priority
Case 1: Objective 1 (passenger security)	1	0.333	0.25	0.122
Case 2: Objective 2 (modernisation)	3	1	0.5	0.32
Case 3: Objective 3 (passenger security and modernisation)	4	2	1	0.558
Consistency ratio	0.016			

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Significantly, the experts modified the element D8 E2 from "Augmentation of Strength of RPF" to "Capacity Augmentation of RPF" because it was realized that mere increase in the strength is not leading to the desired results. The need of the times to come is multi-skilling and enabling the force personnel by providing state of the art technology. Even giving more legal powers to RPF is seen as a way of enhancing the capability of RPF.

	Case 1: Objective 1 (passenger security)	Case 2: Objective 2 (modernisation)	Case 3: Objective 3 (passenger security and modernisation)	Priority
Case 1: Objective 1 (passenger security)	1	0.25	0.2	0.097
Case 2: Objective 2 (modernisation)	4	1	0.5	0.333
Case 3: Objective 3 (passenger security and modernisation)	5	2	1	0.57
Consistency ratio	0.021			

Table XII. Comparison of alternative options on Criterion 5 (technological advancement)

	Case 1: Objective 1 (passenger security)	Case 2: Objective 2 (modernisation)	Case 3: Objective 3 (passenger security and modernisation)	Priority
Case 1: Objective 1 (passenger security)	1	0.333	0.333	0.14
Case 2 : Objective 2 (modernisation)	3	1	0.5	0.333
Case 3 : Objective 3 (passenger security and modernisation)	3	2	1	0.528
Consistency ratio	0.046			

Table XIII. Comparison of alternative options on Criterion 6 (scalability)

	Case 1: Objective 1 (passenger security)	Case 2: Objective 2 (modernisation)	Case 3: Objective 3 (passenger security and modernisation)	Priority
Case 1: Objective 1 (passenger security)	1	0.5	0.333	0.169
Case 2 : Objective 2 (modernisation)	2	1	1	0.387
Case 3 : Objective 3 (passenger security and modernisation)	3	1	1	0.443
Consistency ratio	0.016			

Table XIV. Comparison of alternative options on Criterion 7 (bureaucratic and political interference)

Alternatives	Priority
Case 1: Objective 1 (passenger security)	0.238
Case 2: Objective 2 (modernisation)	0.295
Case 3: Objective 3 (passenger security and modernisation)	0.467

Table XV. Global priorities

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Thus, emphasis should be on augmenting manpower capability, rather than manpower *per se*. This can be achieved by proper multi-skilling. The merger of the different law enforcement agencies into one and enlarging the scope of that single agency for railway security with a collaborative interface with the local police can improve manpower utilisation.

Table XVI. Consistency ratios of criteria for alternative option profiles

S. No.	Criteria	Consistency ratio
1	Feasibility	0.01
2	Effectiveness	0.02
3	Overall cost	0.02
4	Reliability	0.02
5	Technological advancement	0.02
6	Scalability	0.05
7	Bureaucratic and political interference	0.02

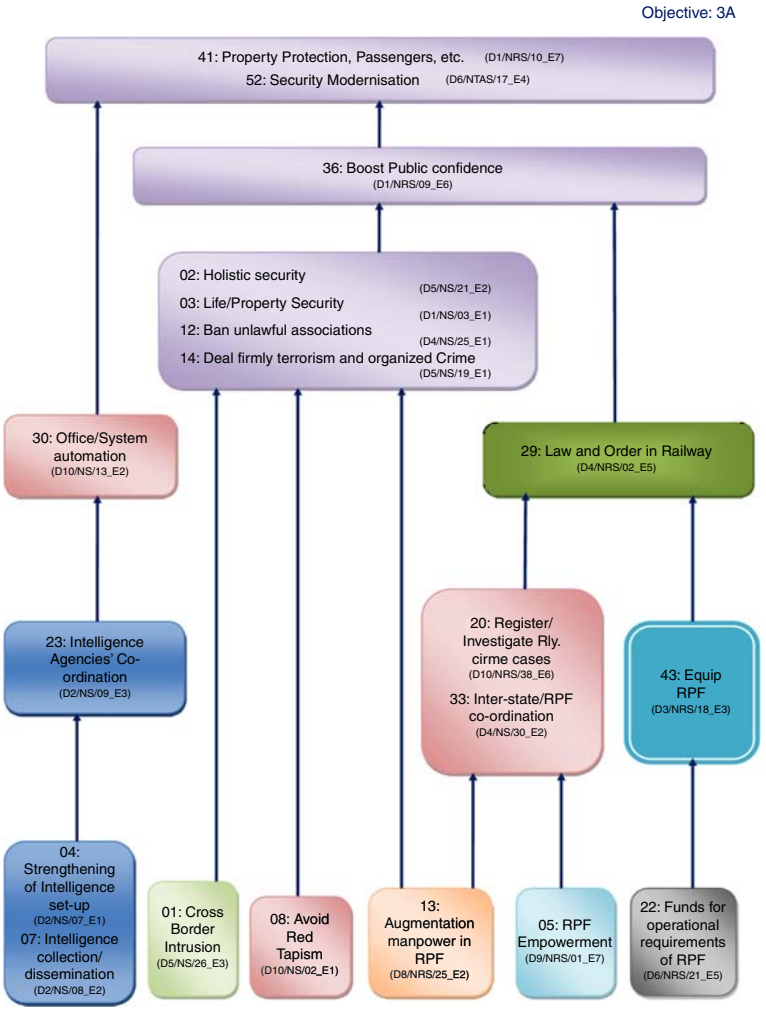


Figure 8. Option profile synthesised by the experts after discussions and deliberations for maximum efficiency of IRSS

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Apart from that, necessary technological aids may be placed at the command of such an agency to make it effective. This kind of arrangement can be tried out in a pilot study to get a practical idea and a field view for evaluation and decision by policy makers accordingly.

The present research can be extended in many important ways. Interpretive structural models for different contextual relationships can be developed and used for formulating and implementing customised security policy, depending upon the local concerns of the system. Policy elements and the ISM structure obtained in this research can be utilised for system dynamic modelling of IRSS. A pilot study can be done to implement the recommendations made in this study. The results can be evaluated using relevant tools to assess the change in the status of IRSS, and decision for further policy planning may be taken accordingly.

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